

Online Appendix:

Predicting Inflation Contagion: A Spatio-Temporal Graph Neural Network Approach to Global Trade

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1 Hyperparameter Grid Search

Table 1 reports validation RMSE for all 18 configurations searched in Milestone 7. The selected model uses learning rate 5×10^{-4} , dropout 0.2, and two GCN layers (lowest validation RMSE 1.252).

Table 1: Hyperparameter grid search on validation RMSE (2019Q1–2020Q4).

Learning rate	Dropout	GCN layers	Val RMSE
0.0005	0.2	2	1.252
0.0005	0.3	2	1.252
0.0005	0.4	2	1.252
0.0010	0.2	2	1.253
0.0010	0.3	2	1.253
0.0010	0.4	2	1.253
0.0005	0.2	1	1.254
0.0005	0.3	1	1.254
0.0005	0.4	1	1.254
0.0010	0.2	1	1.256
0.0010	0.3	1	1.256
0.0010	0.4	1	1.256
0.0030	0.2	1	1.257
0.0030	0.3	1	1.257
0.0030	0.4	1	1.257

Learning rate	Dropout	GCN layers	Val RMSE
0.0030	0.2	2	1.258
0.0030	0.3	2	1.258
0.0030	0.4	2	1.258

2 Supplementary Explainability Figures

2.1 Germany: Domestic vs. Trade Attribution (2020Q2)

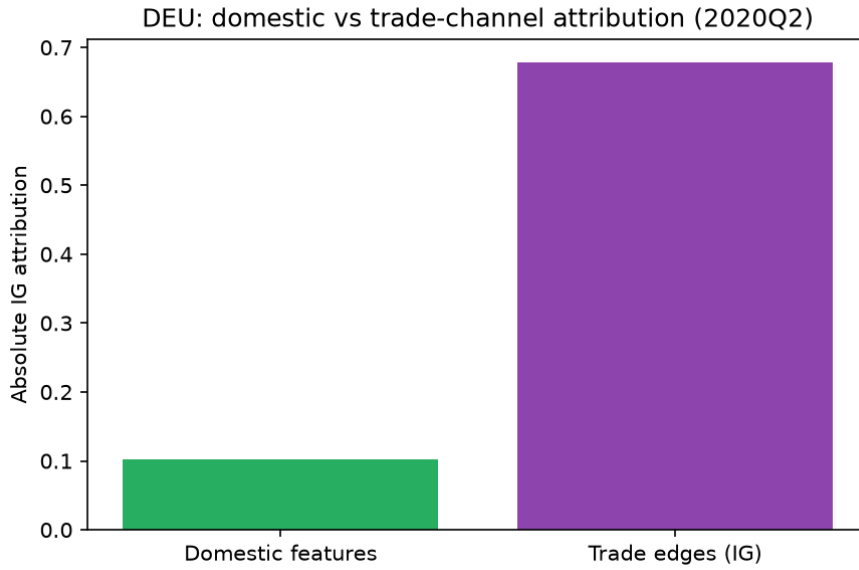


Figure 1: Decomposition of Integrated Gradients attributions for Germany during the COVID supply-chain period (2020Q2).

2.2 Germany: Domestic vs. Trade Attribution (2022Q2)

2.3 Indonesia: Signed Partner Attributions (2022Q2)

2.4 Indonesia: Domestic vs. Trade Attribution (2022Q2)

3 Country List

Table 2 lists the 23 ISO3 codes in the estimation panel.

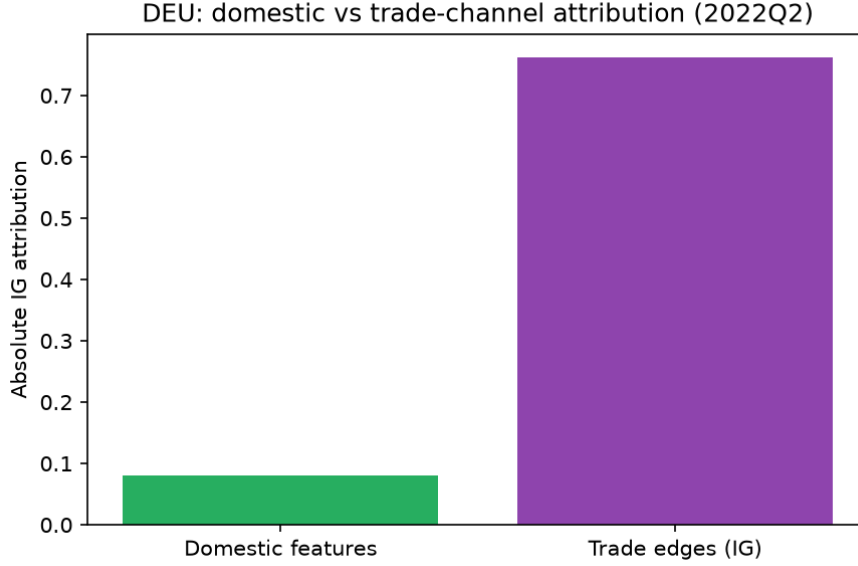


Figure 2: Decomposition of Integrated Gradients attributions for Germany during the 2022 energy shock (2022Q2).

Table 2: Estimation panel (23 countries).

ARG	AUS	BRA	CAN	CHE	CHN
DEU	FRA	GBR	IDN	IND	ITA
JPN	KOR	MEX	NLD	RUS	SAU
SGP	TUR	USA	VNM	ZAF	

4 Replication Instructions

Full replication code is provided in `Replication_Code.zip`. Macro data download automatically via `python -m src.download_macro`. CEPII BACI bulk files must be obtained separately from CEPII (several GB; not included in the zip). See `REPLICATION_README.md` inside the zip for the complete pipeline from raw data through explainability figures.

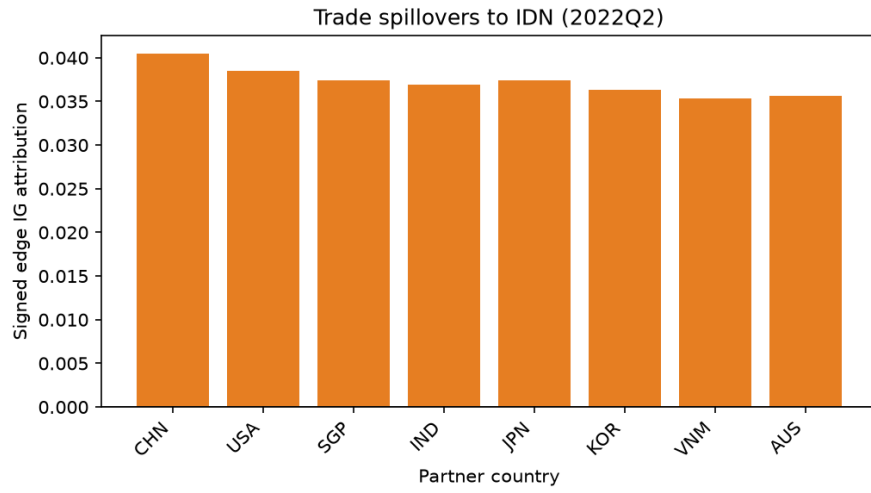


Figure 3: Signed edge attributions for Indonesia's top trade partners (2022Q2).

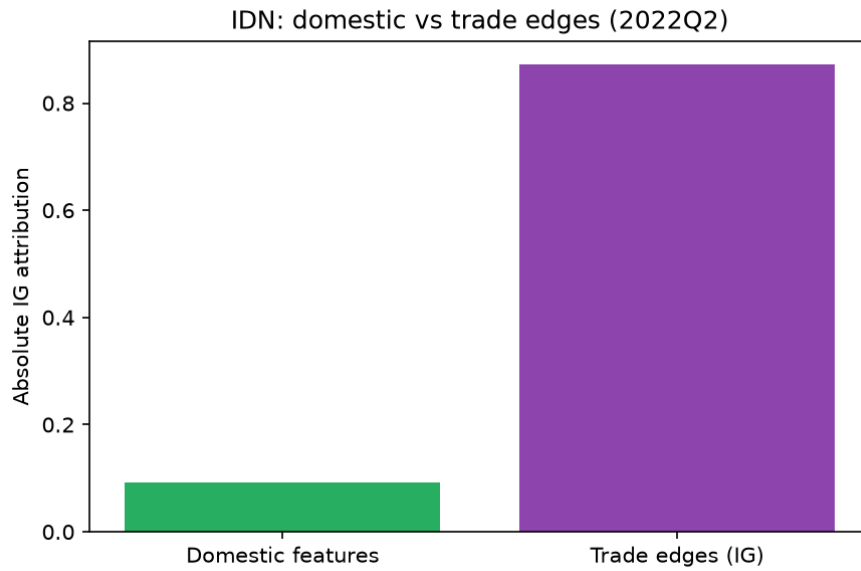


Figure 4: Indonesia trade-versus-domestic attribution ratio (9.44), highest in the panel at 2022Q2.